

Tool #1: Subfinder (The Passive Detective)

What it does: Subfinder is like a detective who looks through public records - certificates, DNS history, search engine caches, etc. It never directly touches the server.

Why use it: Fast, safe (passive), finds subdomains from many sources.

```

--
App: Pass
Host: 200MB
patrick@kali:~$

patrick@kali:~$ subfinder --help
subfinder is a subdomain discovery tool that discovers subdomains for websites by using passive online sources.

USAGE:
  subfinder [flags]

FLAGS:
  -d, --domain string[]  domains to find subdomains for
  -dL, --list string      file containing list of domains for subdomain discovery

SOURCE:
  -s, --sources string[]  specifies sources to use for discovery (i.e. attack,gitlab). Use -s to display all available sources.
  --recursive             use only sources that can handle subdomains recursively rather than both recursive and non-recursive sources
  --all                  use all sources for enumeration (deprecated)
  --exclude-sources string[] sources to exclude from enumeration (i.e. all,attack,gitlab)

FILTER:
  -f, --match string[]  subdomain or list of subdomains to match (file or comma separated)
  -F, --filter string[] subdomain or list of subdomains to filter (file or comma separated)

WORK-LIMIT:
  -cL, --concurrency int  maximum number of HTTP requests to send per second (default)
  -tL, --threads int       maximum number of HTTP requests to send per second (the provided is dependent on your system's hardware) (default=10) (min=1, max=100)
  -tL, --threads int       maximum number of HTTP requests to send per second (the provided is dependent on your system's hardware) (default=10) (min=1, max=100)
  -u, --url string         URL to use for enumeration (i.e. http://)
  --max-time int          maximum time to run in seconds (default=30)
  --max-connections int   maximum number of concurrent connections (default=10)

UPDATE:
  --update               update subfinder to latest version
  --disable-update-check disable automatic subfinder update check

OUTPUT:
  -o, --output string      file to write output to
  --json                 write output in JSON format
  --output-dir string      directory to write output (i.e. dir)
  -oL, --output-dir string include all outputs in the output (i.e. dir)
  --ip                   include host IP in output (i.e. ip)

CONFIGURATION:
  --config string          file config file (default: "/home/patrick/.config/subfinder/config.yml")
  --previous-config string previous config file (default: "/home/patrick/.config/subfinder/previous-config.yml")
  -c string[]              comma separated list of subdomains to use
  -dL string[]             file containing list of domains to use
  --display               display output subdomains only
  --proxy string           keep proxy to use with subfinder
  --no-proxy              exclude ip from the list of domains

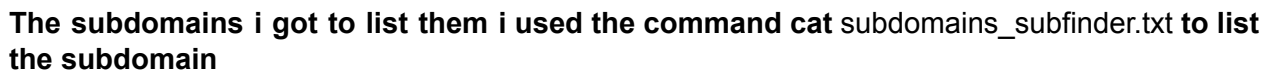
EXAMPLE:
  --help                  show help subfinder is output
  --recursive             show recursive subfinder
  --all                  show recursive subfinder
  -d, --domain            domain name to use
  -s, --sources           list all available sources
  --filter                filter source subdomains

DESCRIPTION:

```

Subfinder -h to help me know how to use the tool

Command that i found to work with to get the subdomain
subfinder -d africahackon.com -o subdomains_subfinder.txt -v



What it does: Dnsenum actually queries the DNS servers directly. It asks what subdomains do you know about?" This is more active and finds different subdomains than Subfinder.

Why use it: Uses a different method, finds subdomains Subfinder might miss.

[illegible]

Command that i used `dnsenum africahackon.com -o dnsenum_output.txt`

[illegible]

What it finds:

[illegible]

Tool #3: Anew

What it does: Now we have subdomains from two different tools. Some will be the same. Anew removes the duplicates and keeps only unique ones.

Why use it: Professional data cleaning. Shows you understand deduplication.

Here's what happens:

Before Anew:

```
[josheta@lvia ~]$ cat /dev/null > /etc/passwd
```

```
# cat /dev/null > /etc/passwd
```

```
africahackon.com  
AT1.ASPMX.L.GOOGLE.COM  
AT2.ASPMX.L.GOOGLE.COM  
AT3.ASPMX.L.GOOGLE.COM  
AT4.ASPMX.L.GOOGLE.COM  
ASPMX.L.GOOGLE.COM  
ftp.africahackos.com  
mail.africahackos.com  
members.africahackos.com  
ml.speccorp.com  
ns.speccorp.com  
ns1.speccorp.com  
ns2.speccorp.com  
ns3.speccorp.com  
webmail.africahackos.com  
www.africahackos.com
```

After Anew:

[illegible]

Run it:

```
# Combine both files and remove duplicates
```

```
cat subdomains_subfinder.txt dnsenum_subdomains.txt | anew final_subdomains.txt
```

Compare the results

Used bashscript to compare the results

[illegible]

[illegible][illegible]

Subfinder found: 62

Dnsenum found: 15

Unique (after Anew): 42

This is the key part:

Out of everything both tools found, **42 subdomains were truly unique.**

PART 2: CHECKING IF THEY'RE ALIVE (The Verification)

Tool #4: Httpx to check their status ALive and dead subdomains

What it does: Httpx sends a request to each subdomain and checks: "Are you there? What status code do you send back?"

Why use it: Separates LIVE subdomains (servers responding) from DEAD ones (no response/offline).

Run it:

```
cat final_subdomains.txt | httpx -sc -title -o httpx_results.txt -v
```

[illegible]

After it finishes:

```
cat httpx results.txt
```



```

- python3liveupdate.py
- curl -s https://www.afrihack.com
https://www.afrihackathon.com [403] [403 Forbidden]
https://www.afrihackathon.com [200] [Forwarded Login]
https://www.afrihackathon.com [403] [403 Forbidden]
https://www.afrihackathon.com [301] [301 Moved Permanently]
https://www.afrihackathon.com [403] [403 Forbidden]
https://www.afrihackathon.com [301] [301 Moved Permanently]
https://www.afrihackathon.com [403]
https://www.afrihackathon.com [301] [Forwarded Login]
https://www.afrihackathon.com [403] [403 Forbidden]
https://www.afrihackathon.com [301] [301 Moved Permanently]
https://www.afrihackathon.com [403]
https://www.afrihackathon.com [403] [403 Forbidden]
https://www.afrihackathon.com [403]
https://www.afrihackathon.com [301] [301 Moved Permanently]
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https://www.afrihackathon.com [301] [Cybersecurity Means Summit]
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https://www.afrihackathon.com [301] [AFriHackathon Academy - The Best Hands-On Trainings, expert guidance, and mentorship to advance your career in Cyber Security]
https://www.afrihackathon.com [301] [AFriHackathon - Nurturing Cybersecurity Excellence in Africa]
https://www.afrihackathon.com [301] [AFriHackathon Talents - The Best Cybersecurity Talents]
https://www.afrihackathon.com [301] [AFriHackathon Events - Africa's Cyber Foundation Is One Platform.]

```

Separating LIVE from DEAD from the results of httpx_results.txt

I used the bash script to do the separation of the alive and the dead



Extract LIVE and dead subdomains



Found 25 alive subdomains and 0 dead subdomains

PART 3: CHECKING FOR FIREWALLS (The Security Analysis)

A firewall/WAF (Web Application Firewall) It checks every request and decides if it's safe.

We use Nuclei to scan

Tool 5 WAFwoof

Based on the alive subdomains



We are going to use the wafw00f to scan the alive subdomains that are there I used bashscript to do it since they were many



And this were the results of the WAF

```
(packetelvis@Elvis)-[~]  
$ cat waf_results.txt
```

```
    ? ,. ( . ) . "  
    __ ?? (" ) )' ,') . ( "  
( __()"; ??? .; ) ' (( (" ) ;(, (( ( ; ) " )")
```

```

/,___ / _", ,'_.,.)_(.,( . )_-' )_') ( . _..(' )
\\ \\ |___|___|___|___|___|___|___|___|___|___|

```

~ WAFW00F : v2.3.1 ~
~ Sniffing Web Application Firewalls since 2014 ~

[*] Checking <https://webdisk.africahackon.com>
[+] Generic Detection results:
[-] No WAF detected by the generic detection
[~] Number of requests: 7

```

_____  

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( Woof! )  

\___/   

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\ ( _ ) ) / | \ |__|

```

~ WAFW00F : v2.3.1 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://www.new.africahackon.com>
[+] The site <https://www.new.africahackon.com> is behind LiteSpeed (LiteSpeed Technologies) WAF.
[~] Number of requests: 2

```

_____  

/\   

( W00f! )  

\___/   

,, __ 404 Hack Not Found  

|`-.__ // __ __  

/'' _/ /_ \ \ \ \   

*===* /\ \_ // 405 Not Allowed  

/ ) __ // \ \   

/ / / / ---` 403 Forbidden  

W ^ \ | / _ \   

\ / _ \ _ 502 Bad Gateway / / \ \ 500 Internal Error  

` _____ ``-` / _ \ _ \

```

The Web Application Firewall Fingerprinting Toolkit

[~] Number of requests: 7

```

/\
( W00f! )
\____/
,, __ 404 Hack Not Found
|'-. _// _ _
/'" _// _\\\\//
*===* /\ _// 405 Not Allowed
| ) _//\\//
/|//---` 403 Forbidden
W` \ | / _ \
` \ / _ _ 502 Bad Gateway //\\ 500 Internal Error `_____`-`-`
/ / \ \

```

The Web Application Firewall Fingerprinting Toolkit

[~] Number of requests: 7

[illegible]

~ Sniffing Web Application Firewalls since 2014 ~

[~] Number of requests: 7

```

      _____
     /\
    ( W00f! )
     \_____/
      „__ 404 Hack Not Found
     |`-.__//__ __
     /" _//_\\//
    *==* /\_// 405 Not Allowed
     /)_//\
  // / /---` 403 Forbidden
  W` \ | / _ \
  \/_\_ 502 Bad Gateway // \ \ 500 Internal Error
  `____`-`-` /_/_\

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://www.members.africahackon.com>

[+] The site <https://www.members.africahackon.com> is behind LiteSpeed (LiteSpeed Technologies) WAF.

[~] Number of requests: 2

```

      _____
     /\
    ( Woof! )
     \_____/
      „) (
     .- - _____ ( | |
     ()``; |==|_____ )_|_|
     / (' /\ ( | |
    ( / ) / | \ . | |
    \ ( _ ) ) / | \ | |

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://www.masterclass.africahackon.com>

[+] The site <https://www.masterclass.africahackon.com> is behind CacheWall (Varnish) WAF.

[~] Number of requests: 2

```

      ? „ ( . ) . "
     __ ?? ( " ) ' , ' ) . ( `

```

(__()" ; ??? ;) ' ((") ; (((;) "))"
/,__ /' _ " , , _ ' _ ,) _ (, , (.) _ _ ') _ ') (. _ , (')
\\ \ | | | | | | | | | | | | | | | |

~ WAFW00F : v2.3.1 ~
~ Sniffing Web Application Firewalls since 2014 ~

[*] Checking <https://new.africahackon.com>
[+] The site <https://new.africahackon.com> is behind LiteSpeed (LiteSpeed Technologies) WAF.
[~] Number of requests: 2

? ,. (.) . "
__ ?? (")) ' , ') . (^ "
(__()" ; ??? ;) ' ((") ; (((;) "))"
/,__ /' _ " , , _ ' _ ,) _ (, , (.) _ _ ') _ ') (. _ , (')
\\ \ | | | | | | | | | | | | | | | |

~ WAFW00F : v2.3.1 ~
~ Sniffing Web Application Firewalls since 2014 ~

[*] Checking <https://masterclass.africahackon.com>
[+] The site <https://masterclass.africahackon.com> is behind CacheWall (Varnish) WAF.
[~] Number of requests: 2

? ,. (.) . "
__ ?? (")) ' , ') . (^ "
(__()" ; ??? ;) ' ((") ; (((;) "))"
/,__ /' _ " , , _ ' _ ,) _ (, , (.) _ _ ') _ ') (. _ , (')
\\ \ | | | | | | | | | | | | | | | |

~ WAFW00F : v2.3.1 ~
~ Sniffing Web Application Firewalls since 2014 ~

[*] Checking <https://ftp.africahackon.com>
[+] The site <https://ftp.africahackon.com> is behind LiteSpeed (LiteSpeed Technologies) WAF.
[~] Number of requests: 2

/\
(W00f!)
_ \ /

```

    „ __ 404 Hack Not Found
    |'-.__// __ __
    /" _//_//\\//
    *===* /\\_// 405 Not Allowed
    /)_//\\
    / / /---` 403 Forbidden
    W`\\|/_\\
    `/_/_ 502 Bad Gateway //\\ 500 Internal Error
    `_____`-`-`/_/_\\

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://members.africahackon.com>

[+] The site <https://members.africahackon.com> is behind LiteSpeed (LiteSpeed Technologies) WAF.

[~] Number of requests: 2

? , (.) . "
 _ ?? (")) ' , ') . (' "
 (_ () " ; ??? . ;) ' ((") ; (, ((;) ") ")
 I , _ _ I _ " , , _ ' _ ,) _ (, (.) _ ') _ ') (. _ (')
 \ \ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ |

~ WAFW00F : v2.3.1 ~

~ Sniffing Web Application Firewalls since 2014 ~

```
[*] Checking https://cpcalendars.africahackon.com
```

[+] Generic Detection results:

[-] No WAF detected by the generic detection

[~] Number of requests: 7

```

      _____
      /\
      ( Woof! )
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      ∴ - _____ ( |__|
      ()``; |==|_____ ) .|__|
      / (' \/\ ( |__|
      (/) / \ \ . |__|
      \(_)_ ) / \ \ |__|

```


~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://lms.africahackon.com>

[+] Generic Detection results:

[-] No WAF detected by the generic detection

[~] Number of requests: 6

```

      ? ,. ( . ) . "
    __ ?? ( " ) ' , ' ) . ( ' "
  ( __ () ^ ; ??? . ; ) ' ( ( " ) ; ( , ( ( ; ) " ) " )
/ , __ / _ " , , _ ' _ , ) _ ( , , ( . ) _ ' ) _ ) ( . _ . ( ' )
\\ \\ | _ _ | _ _ | _ _ | _ _ | _ _ | _ _ | _ _ | _ _ | _ _ |

```

~ WAFW00F : v2.3.1 ~

~ Sniffing Web Application Firewalls since 2014 ~

[*] Checking <https://webmail.africahackon.com>

[+] Generic Detection results:

[*] The site <https://webmail.africahackon.com> seems to be behind a WAF or some sort of security solution

[~] Reason: The server header is different when an attack is detected. The server header for a normal response is "LiteSpeed", while the server header a response to an attack is "",

[~] Number of requests: 6

```

      _____
      / \
      ( W00f! )
      \ _____ /
      , , __ 404 Hack Not Found
      | ^ . _ // _ _ _
      / " _ / / _ \ \ \ \
      * == * / \ _ // 405 Not Allowed
      / ) _ // \ \
      / / / --- ` 403 Forbidden
      W ^ \ | / _ \
      ` \ / _ _ 502 Bad Gateway / / \ \ 500 Internal Error
      ` _____ ` ^ ^ / _ \ _ \

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://www.africahackon.com

```
_____  
/\   
( Woof! )  
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  „ ) ( _  
  .- - _____ ( | _ |  
  ()``; |==|_____ ) .)| _ |  
  / (' / \ ( | _ |  
  ( / ) / | \ . | _ |  
  \ ( _ ) ) / | \ | _ |
```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://www.talent.africahackon.com

[+] Generic Detection results:

[*] The site https://www.talent.africahackon.com seems to be behind a WAF or some sort of security solution

[~] Reason: Blocking is being done at connection/packet level.

[~] Number of requests: 1

```
_____  
/\   
( W00f! )  
 \____/   
  „ __ 404 Hack Not Found  
  |`-.__ // __ __  
  /" _//_//\ \ \ \   
  *==* /\ \ _// 405 Not Allowed  
  / ) _// \ \   
  /| / /---` 403 Forbidden  
  W` \ | / _ \   
  ` \ / _ \ 502 Bad Gateway / / \ \ 500 Internal Error  
  ` _____ ``-` / / \ _ \
```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://mail.africahackon.com

[+] Generic Detection results:

[*] The site <https://mail.africahackon.com> seems to be behind a WAF or some sort of security solution

[~] Reason: Blocking is being done at connection/packet level.

[~] Number of requests: 2

```

      _____
     / \
    ( W00f! )
     \____/
      „__ 404 Hack Not Found
     |`-.__//__ __
     /" _//_\\ \ \ \ \
    *===* /\_\_// 405 Not Allowed
     /)_// \ \
  /| / /---` 403 Forbidden
 W` \ | / _ \
 \/_\_ 502 Bad Gateway / / \ \ 500 Internal Error
 `____`-`-` /_/_ \ \

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://www.summit.africahackon.com>

[+] Generic Detection results:

[*] The site <https://www.summit.africahackon.com> seems to be behind a WAF or some sort of security solution

[~] Reason: Blocking is being done at connection/packet level.

[~] Number of requests: 5

```

      _____
     / \
    ( W00f! )
     \____/
      „__ 404 Hack Not Found
     |`-.__//__ __
     /" _//_\\ \ \ \ \
    *===* /\_\_// 405 Not Allowed
     /)_// \ \
  /| / /---` 403 Forbidden
 W` \ | / _ \
 \/_\_ 502 Bad Gateway / / \ \ 500 Internal Error
 `____`-`-` /_/_ \ \

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://summit.africahackon.com>

```
_____  
/\   
( W00f! )  
 \____/  
,, __ 404 Hack Not Found  
|`-.__// __ __  
/'' _//_//\ \//  
*==*_//\ _// 405 Not Allowed  
/)_ _//\ \/  
/| / /---` 403 Forbidden  
\\` \ | / _ \  
` \ / _ \ 502 Bad Gateway / / \ \ 500 Internal Error  
` _____ ``-` / _ \ _ \
```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://africahackon.com>

[+] Generic Detection results:

[-] No WAF detected by the generic detection

[~] Number of requests: 7

```
? ,. ( . ) . "  
__ ?? (" ) ' , ' ) . ( "  
( __ ( ) " ; ??? . ; ) ' ( ( " ) ; ( , ( ( ; ) " ) "  
/ , __ / _ " , , _ ' _ , ) _ ( , ( . ) _ _ ' ) _ ' ) ( . _ _ ( ' )  
\\ \\ | _ | _ | _ | _ | _ | _ | _ | _ | _ | _ |
```

~ WAFW00F : v2.3.1 ~

~ Sniffing Web Application Firewalls since 2014 ~

[*] Checking <https://www.events.africahackon.com>

[+] Generic Detection results:

[*] The site <https://www.events.africahackon.com> seems to be behind a WAF or some sort of security solution

[~] Reason: Blocking is being done at connection/packet level.

[~] Number of requests: 6

```

/\
( W00f! )
\____/
,, __ 404 Hack Not Found
|`-.__//__ __
/"" _//_\\//
*===*/\\_// 405 Not Allowed
/)_//\\
/| / /---` 403 Forbidden
W`\\|/_\
`/_\\_ 502 Bad Gateway //\\ 500 Internal Error
`_____``-`/_/_\

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://www.academy.africahackon.com>

[+] Generic Detection results:

[-] No WAF detected by the generic detection

[~] Number of requests: 7

```

_____
/\
( W00f! )
\____/
,, __ 404 Hack Not Found
|`-.__//__ __
/"" _//_\\//
*===*/\\_// 405 Not Allowed
/)_//\\
/| / /---` 403 Forbidden
W`\\|/_\
`/_\\_ 502 Bad Gateway //\\ 500 Internal Error
`_____``-`/_/_\

```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://talent.africahackon.com>

[+] Generic Detection results:

[*] The site <https://talent.africahackon.com> seems to be behind a WAF or some sort of security solution

[~] Reason: Blocking is being done at connection/packet level.

[~] Number of requests: 2

```
_____  
/\   
( W00f! )  
 \____/  
„__ 404 Hack Not Found  
|`-.__//__ __  
/\"_//_//\ \/  
*==* /\_// 405 Not Allowed / )__// \/  
/| / /---` 403 Forbidden  
W` \ | / _ \  
` \ / _ _ 502 Bad Gateway / / \ \ 500 Internal Error `____`-`  
/_// \ _\
```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://events.africahackon.com>

```
_____  
/\   
( Woof! )  
 \____/  
„) ( _  
.-. - _____ ( | _ |  
( ) ``; |==|_____ ) .)| _ |  
/ ( ' / | \ ( | _ |  
( / ) / | \ . | _ |  
 \ ( _ ) ) / | \ | _ |
```

~ WAFW00F : v2.3.1 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking <https://academy.africahackon.com>

[+] Generic Detection results:

[-] No WAF detected by the generic detection

[~] Number of requests: 7